CAR DETECTION

Car Detection using HOG Features + SVM Classifier + Sliding Window

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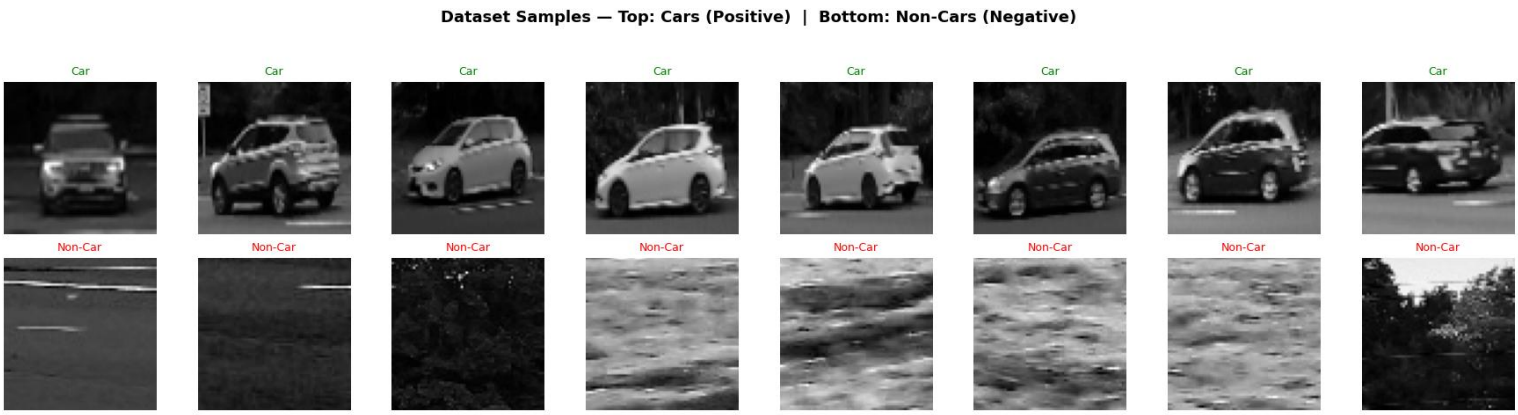
Objective

This project implements a classical car detection pipeline using HOG features, a Linear SVM classifier, and a sliding window approach — without any deep learning.

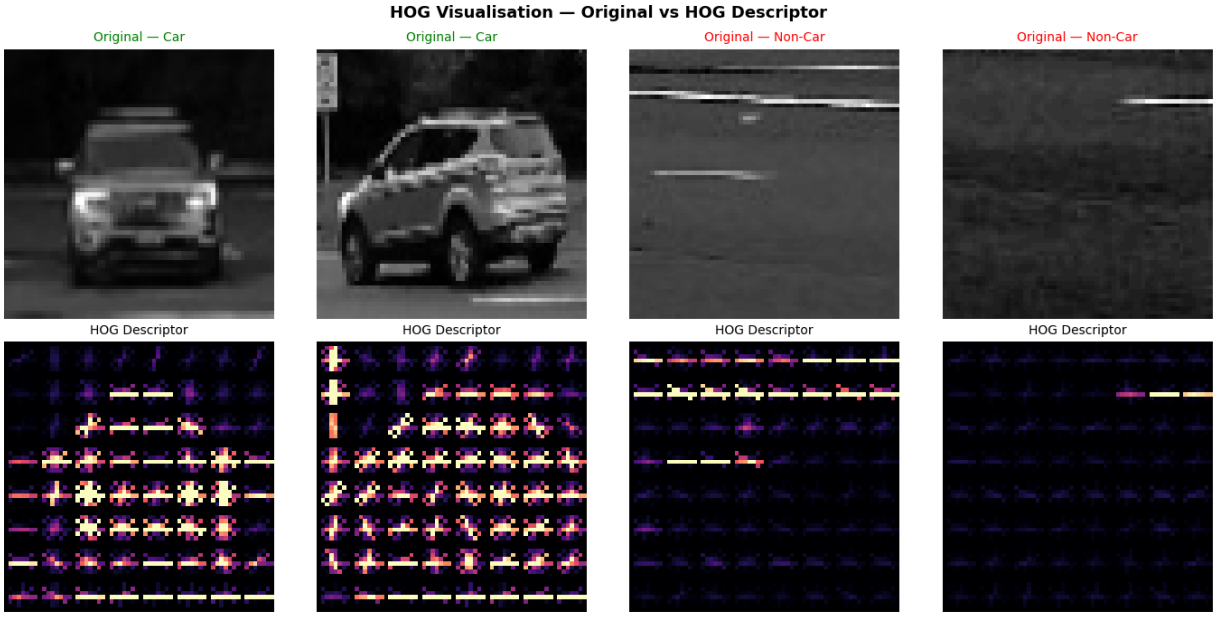
Dataset Preparation

The dataset consists of 1,001 training images with 559 annotated bounding boxes across 355 unique images. Positive samples (cars) were extracted by cropping ground-truth bounding boxes from each annotated image, yielding 559 car patches. Negative samples (non-cars) were collected by randomly sampling 64×64 windows from each image, with IoU filtering (threshold < 0.1) to ensure no negative patch accidentally overlapped a car. Three negatives were sampled per image, producing ~3,003 negative patches total.



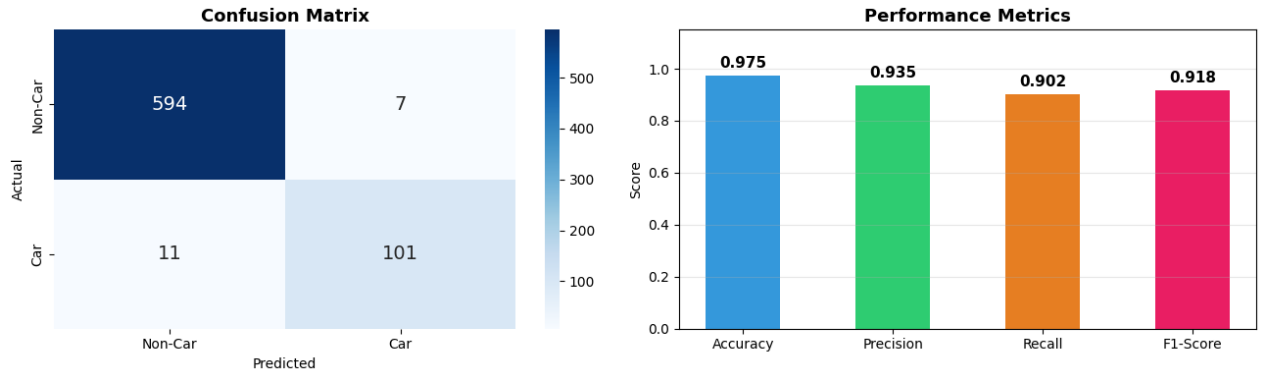
Feature Extraction

Each patch was resized to 64×64 pixels and passed through a HOG descriptor with 9 orientation bins, 8×8-pixel cells, and 2×2 cell blocks with L2-Hys normalization, producing a 1,764-dimensional feature vector per patch that captures edge structure and shape while remaining robust to illumination changes.



Classifier Training & Results

A Linear SVM (C=0.01) was trained on an 80/20 stratified split with StandardScaler normalization. The classifier achieved the following on the validation set (713 samples):



The confusion matrix shows 594 true negatives and 101 true positives, with only 7 false positives and 11 false negatives — confirming the classifier generalizes well across both classes.

Sliding Window Detection

The detector scanned each image at 6 scales [1.0, 1.25, 1.5, 1.75, 2.0, 2.5] with a stride of 8 pixels. A confidence threshold of 1.86 was applied to filter weak detections, followed by Non-Maximum Suppression (NMS) at IoU 0.15 to eliminate duplicate boxes around the same car.

Image	Detections	Note
vid_5_26620	1	Correct
vid_5_26640	1	Correct
vid_5_26660	0	Missed — motion blur
vid_5_26680	1	Correct
vid_5_26700	0	Missed — motion blur, low contrast
vid_5_26720	1	Correct
vid_5_26740	0	Missed — motion blur
vid_5_26760	1	Correct
vid_5_26780	1	Correct
vid_5_26800	2	Correct
vid_5_26820	2	Correct

The detector successfully localized cars in 8 out of 11 test frames.

Limitations

- Motion blur degrades HOG features since they rely on sharp edges
- Cars at extreme scales or unusual orientations may be missed
- The sliding window search is computationally expensive at inference time